Building Materials

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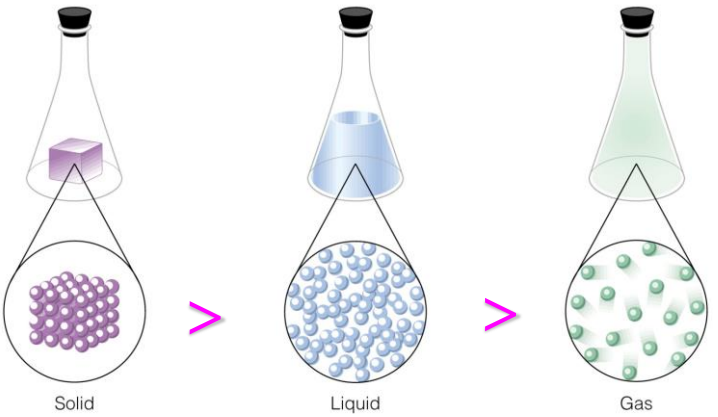
Chapter One Introduction
Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
Chapter Five Asphalt
Chapter Six Masonry
Chapter Seven Wood

Chapter Two: The Fundamentals of Materials

1. Physical Architecture of Materials
2. Behavior of Materials Under Stress
3. Surface Properties
4. Failure and Fracture

2.1 Physical Architecture of Materials

Three states of matter.

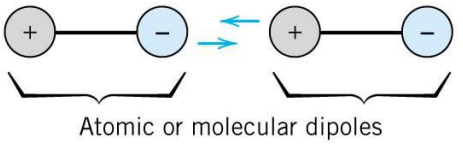


The state of a substance at room temperature depends on the **strength of the attractions between atoms**.

2.1 Physical Architecture of Materials

Bonding Between Atoms

- Formation of bonds: attraction between oppositely charged poles



Atomic or molecular dipoles

- Strength of Bonds

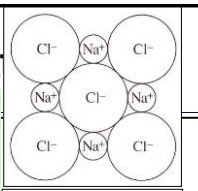
- **Primary bonds: ionic, covalent and metallic bonding**
  - strong bonds with bonding energy from a few hundred to one thousand kJ/mole
- Secondary bonds: van der Waals bonding
  - weak bonds
  - bonding energy of the order of 10 kJ/mol

2.1 Physical Architecture of Ma

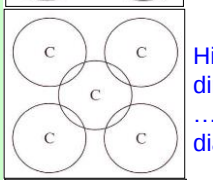
1. Ionic bonds:
2. Covalent bonds:
3. Metallic bonds:
4. Van Der Walls Bonding:

Strong and primary bonds

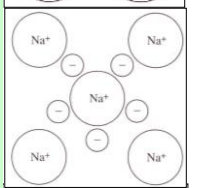
weak bonds



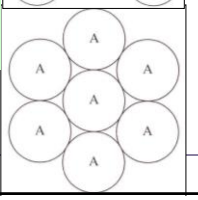
Salt, gypsum



Highly directional … diamond



metal

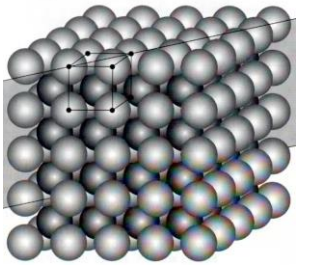


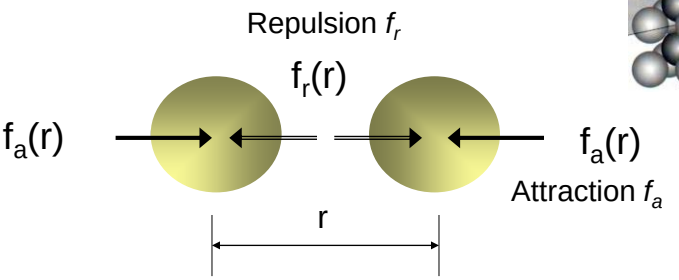
polymer

Which types of bonding will cause high electrical and thermal conductivity?

2.1 Physical Architecture of Materials

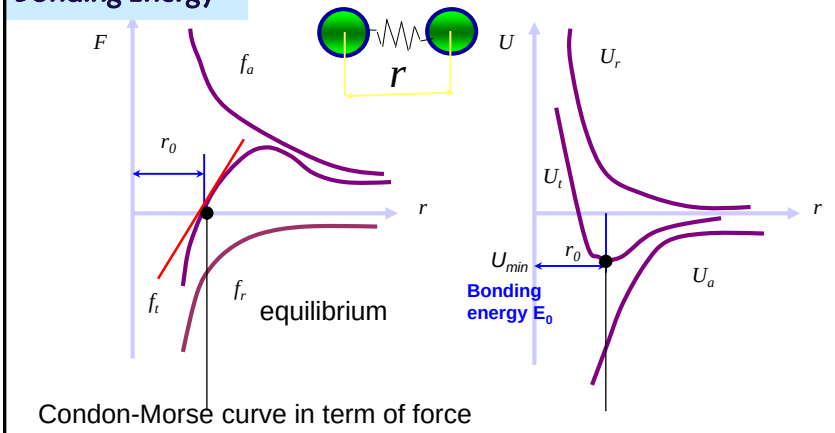
Bonding Energy





2.1 Physical Architecture of Materials

Bonding Energy



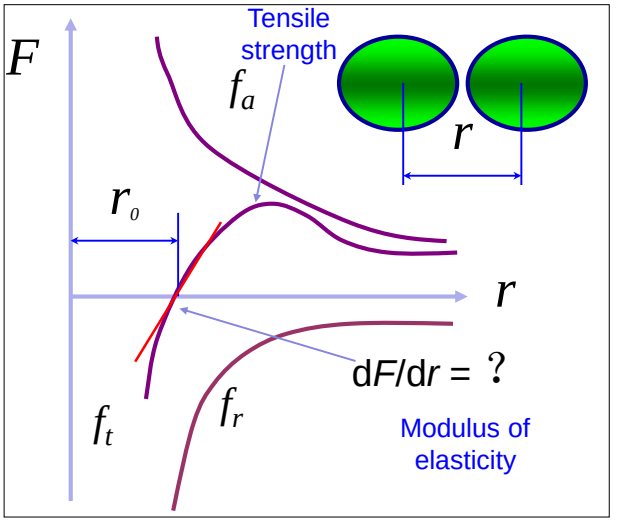
Condon-Morse curve in term of force

$r = r_0$: $|f_a| = |f_r|$, $F = 0$

r_0 : equilibrium distance between atoms

Bonding energy: the energy that would be required to separate these two atoms to an infinite separation.

2.1 Physical Architecture of Materials



Condon-Morse curve in term of force

Strength:

Modulus:

$$\frac{dF}{dr} = E$$

2.1 Physical Architecture of Materials

Colloid:

- Colloid is a highly dispersed phase with a mean particle dimension of less than 1um.
- Each kind of phase (solids, liquids, or gases) can exist in a colloidal form and may be dispersed in either of the other phases.

Tyndall effect: 丁达尔效应



For examples:
jelly、cement
paste、fog

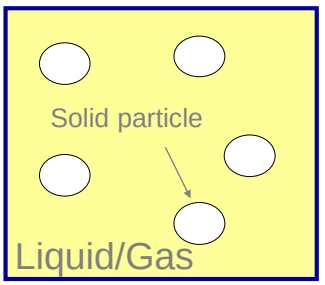


2.1 Physical Architecture of Materials

Colloid:

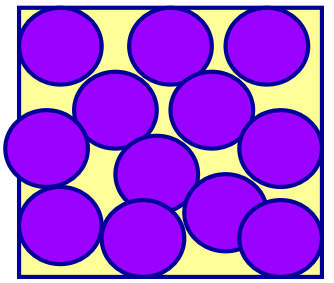
There are two type of colloids: Colloidal sol and Colloidal gel。

Sol (溶胶)



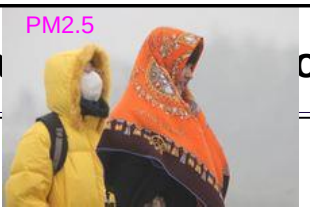
Sol: dispersion of solids in a liquid or gas.

Gel (凝胶)



Gel: sols interact to form a 3-D network, behave as a high viscosity fluid or even a solid.

2.1 Physica of Mat





Colloid:

Examples:

Dispersed Phase	Surrounding Phase	Type	Examples
Gas	Liquid Solid	Foam Solid Foam	Soapy water Air-entrained concrete
Liquid	Gas Liquid	Aerosol Emulsions	Atmospheric smog Milk, emulsified asphlt
Solid	Gas Solid	Smokes Solid	Cement paste

Concrete is gel !

2.1 Physical Architecture of Materials

Colloid: **Thixotropy**

the property of certain gels or fluids that are thick (viscous) under normal conditions, but flow (become thin, less viscous) over time when shaken, agitated, or otherwise stressed.



landslide





The lahar from the 1985 eruption that wiped out the town in Colombia.

2.1 Physical Architecture of Materials

Colloid: **Thixotropy**



Chapter Two: The Fundamentals of Materials

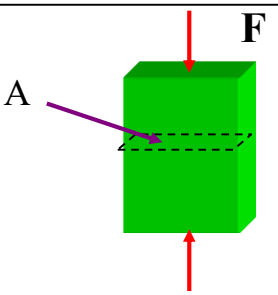
1. Physical Architecture of Materials
2. Behavior of Materials Under Stress
3. Surface Properties
4. Failure and Fracture

2.2 Behavior of Materials under Stress

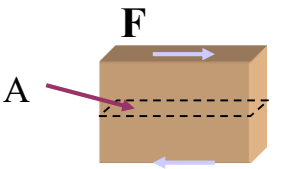
Strain and Stress:

Definitions:

Stress: $\sigma = \frac{F}{A}$



Shear stress: $\tau = \frac{F}{A}$

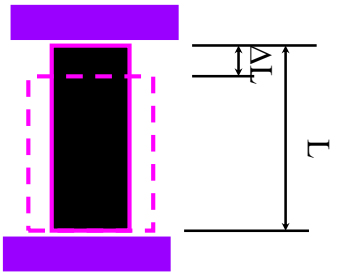


unit : N/m^2 (Pa), N/mm^2 (MPa)

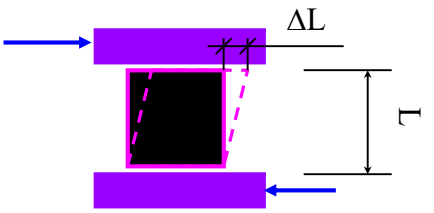
2.2 Behavior of Materials under Stress

Strain and Stress:

Strain: $\varepsilon = \dfrac{\Delta L}{L}$



Shear strain: $\gamma = \dfrac{\Delta L}{L}$



dimensionless

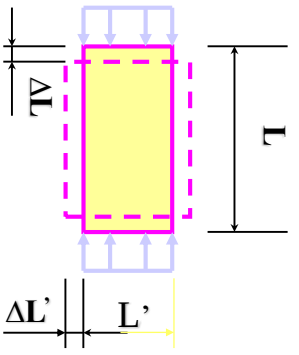
2.2 Behavior of Materials under Stress

Mechanical Properties

axial strain: $\varepsilon = \dfrac{\Delta L}{L}$

transverse strain: $\varepsilon' = \dfrac{\Delta L'}{L'}$

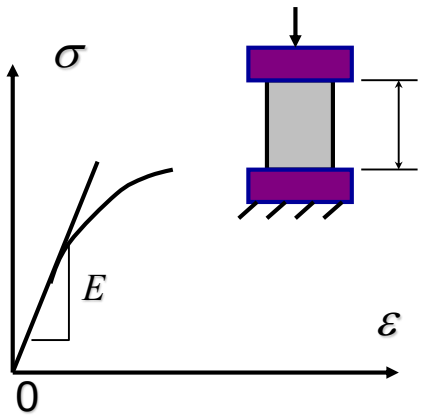
Poisson ratio: $\nu = -\dfrac{\varepsilon'}{\varepsilon}$



ν : dimensionless

2.2 Behavior of Materials under Stress

Young' s Modulus



Young's modulus E

$$\sigma = E\varepsilon$$

Hooke's law

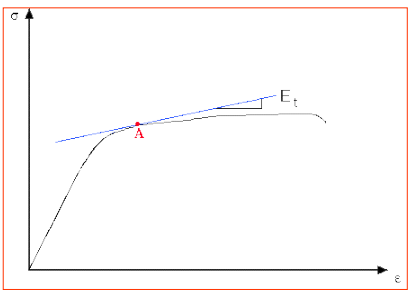
Shear modulus G

$$\tau = G\gamma$$

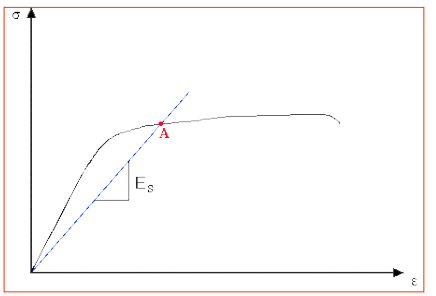
unit : E,G (Pa, MPa, GPa)

2.2 Behavior of Materials under Stress

Tangent and Secant Modulus



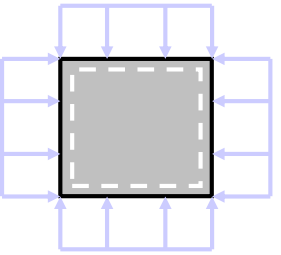
Slope of a line tangent to the stress-strain curve at the point of interest.



Slope of a line from the origin of the stress-strain diagram and intersecting the curve at the point of interest.

2.2 Behavior of Materials under Stress

Bulk Modulus:



pressure　　Volume change

$$P = -K\frac{dV}{V}$$

Volume modulus of elasticity
Or
Bulk modulus

2.2 Behavior of Materials under Stress

Modulus:

Four material parameters: **(*E*, *G*, *K*, *ν*)**

For isotropic materials:

$$G = \frac{E}{2(1+\nu)}$$

$$K = \frac{E}{3(1-2\nu)}$$

2.2 Behavior of Materials under Stress

Rheology

☐ Rheology is deformation and flow of matter under controlled testing conditions .



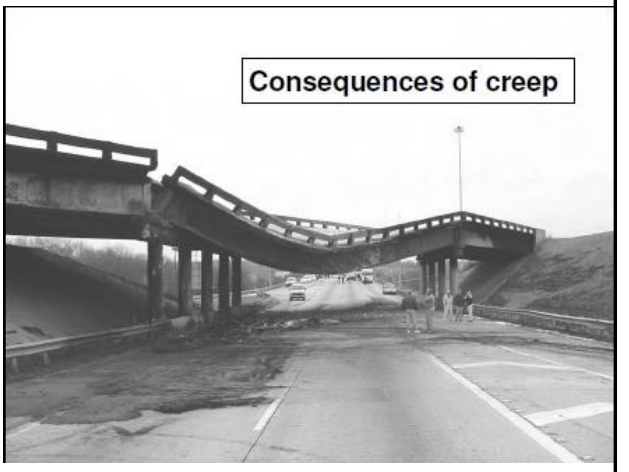
2.2 Behavior of Materials under Stress

Rheology

Definitions:

Creep = deformation occurring under a constant sustained stress

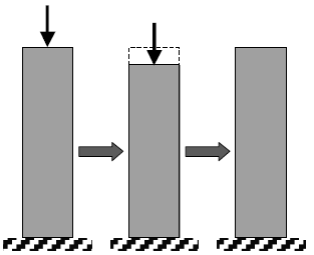
Relaxation = decrease in stress with time under a constant displacement



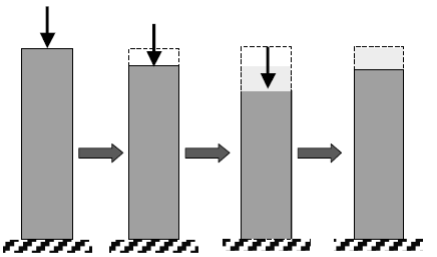
2.2 Behavior of Materials under Stress

Rheology

Creep is the increase in strain under sustained stress (direct strain, flexural strain, etc)



Elastic deformation



Creep

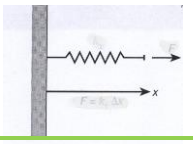
2.2 Behavior of Materials under Stress

Rheology

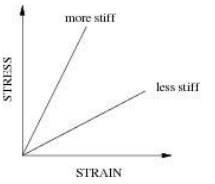
Mechanical analogues

Stress-Strain relations

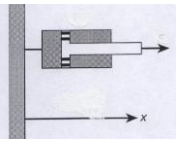
Elastic (spring)



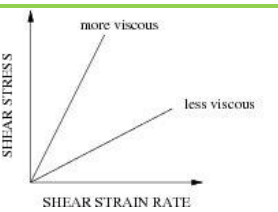
$$\sigma = E\varepsilon$$



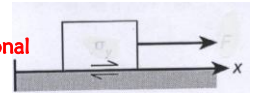
Viscous (dashpot)



$$\sigma = \eta \frac{d\varepsilon}{dt}$$

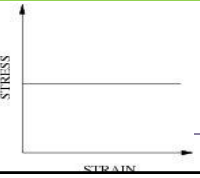


Plastic (sliding frictional element)



$$\text{slip} = 0 \text{ if}$$

$$\sigma < \mu\sigma_w$$



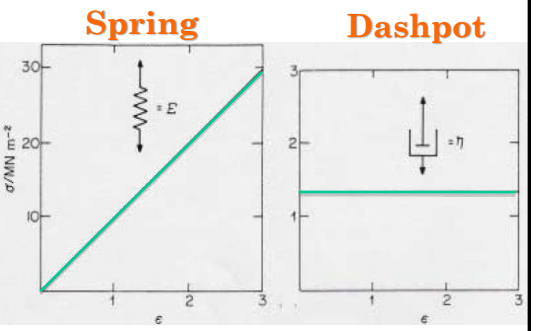
2.2 Behavior of Materials under Stress

Rheology

Springs

- Purely Elastic
- Respond Instantaneously
- Store Energy
- Hook's Law

$$\sigma = E\varepsilon$$



Dashpots

- Purely Viscous
- Slow response
- Dissipate Energy in form of heat
- Newton's Law

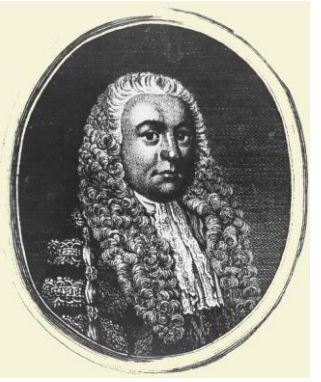
$$\sigma = \eta \frac{d\varepsilon}{dt}$$

η : is viscosity

2.2 Behavior of Materials under Stress

Ideal Solids: Hook's Law 1676



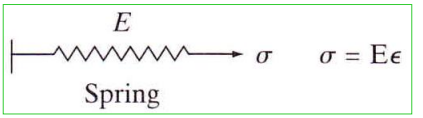


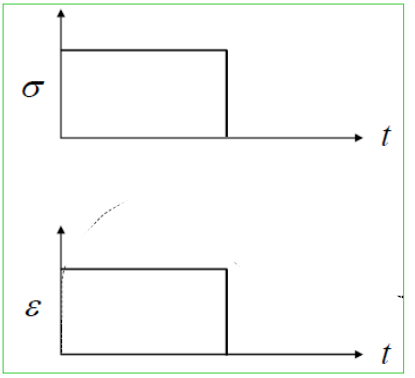
http://www.upscale.utoronto.ca/GeneralInterest/Harrison/Flash/ClassMechanics/HookesLaw/HookesLaw.html

2.2 Behavior of Materials under Stress

Rheology

Hookeian element:





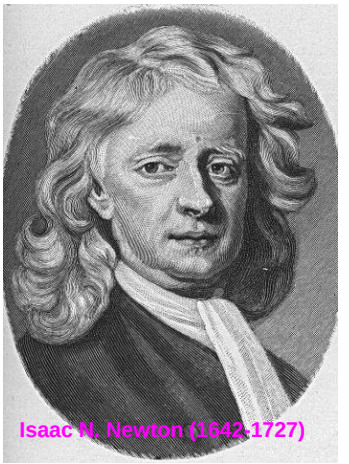
2.2 Behavior of Materials under Stress

Rheology

Viscos dashpot: $\sigma = \eta \frac{d\varepsilon}{dt}$

Ideal Liquids: Newton's Law:

$$\tau_{yx} = \eta \dot{\gamma}$$



Isaac N. Newton (1642-1727)

2.2 Behavior of Materials under Stress

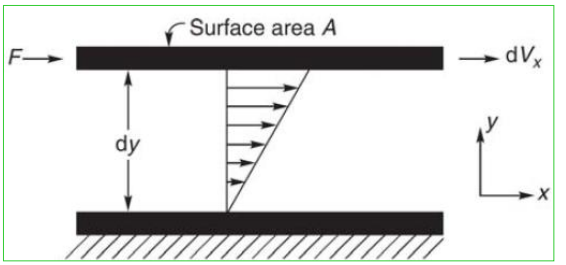
Rheology

Ideal Liquids:
Newton's Law:

$$\frac{F}{A} = \tau_{yx} = \eta \frac{V}{Y}$$

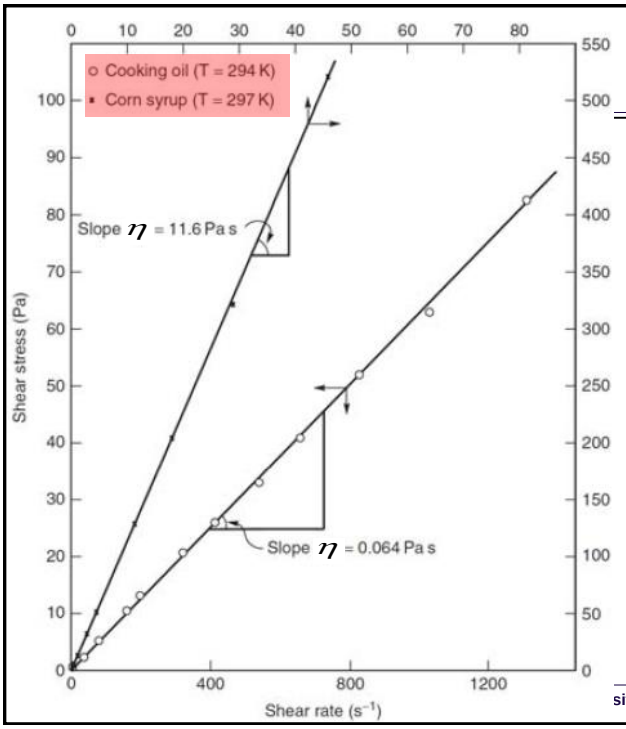
$$\tau_{yx} = \eta \left(\frac{dV}{dy} \right)$$

$$\tau_{yx} = \eta \dot{\gamma}$$



$$\dot{\gamma} = \frac{d\gamma}{dt} = \frac{d\left(dx/dy \right)}{dt} = \frac{d(dx/dt)}{dy} = \left(\frac{dV}{dy} \right)$$

Units of viscosity η is Pa s (Pascal second).



Which one is more viscous?

Typical shear stress–shear rate data for a cooking oil and a corn syrup

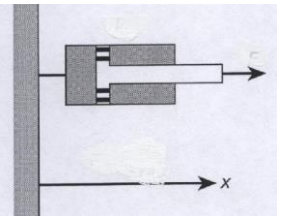
2.2 Behavior of Materials under Stress

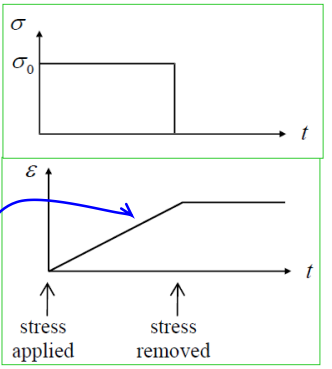
Rheology

Viscos dashpot:

$$\tau_{yx} = \eta \dot{\gamma}$$

$$\sigma = \eta \frac{d\varepsilon}{dt}$$





$$\varepsilon = \frac{\sigma_0}{\eta} t$$

2.2 Behavior of Materials under Stress

Maxwell Model for Viscoelastic Behavior

Developed to explain asphalt
- Assume strains are additive
- Assume stress constant

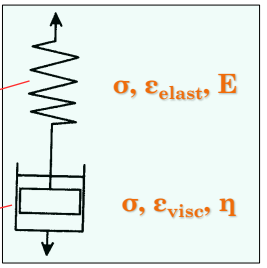
$$\varepsilon_1 = \frac{1}{E}\sigma$$

$$\dot{\varepsilon}_2 = \frac{1}{\eta}\sigma$$

$$\varepsilon = \varepsilon_1 + \varepsilon_2$$

$$\dot{\varepsilon} = \frac{1}{E}\dot{\sigma} + \frac{1}{\eta}\sigma$$

$$\sigma + \frac{\eta}{E}\dot{\sigma} = \eta\dot{\varepsilon}$$

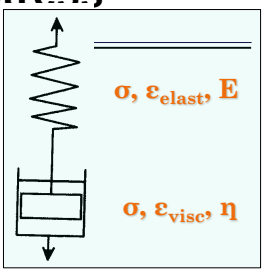




James Clerk Maxwell
1831-1879

2.2 Behavior of Materials under Stress

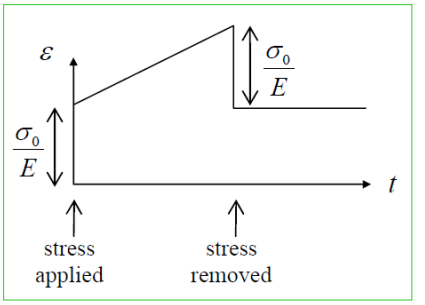
Maxwell Model for Viscoelastic Behavior



What is the response of this model to a load σ_0 ?

$$\dot{\varepsilon} = \frac{\sigma_0}{\eta} \Rightarrow \varepsilon(t) = \frac{\sigma_0}{\eta} t + C$$

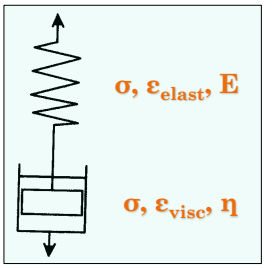
$$\varepsilon(t) = \frac{\sigma_0}{\eta} t + \frac{\sigma_0}{E}$$



creep

2.2 Behavior of Materials under Stress

Maxwell Model for Viscoelastic Behavior



$$\sigma + \frac{\eta}{E}\dot{\sigma} = \eta\dot{\varepsilon}$$

(both terms: 0 when $\sigma = \sigma_0$)

Creep Test: Constant stress: $\sigma = \sigma_0$

$$\varepsilon(t) = \frac{\sigma_0}{E} + \frac{\sigma_0}{\eta} t$$

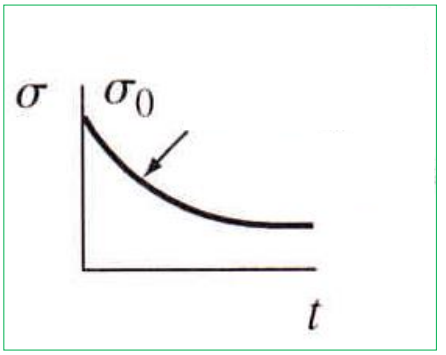
Relaxation Test: Constant strain: $\varepsilon = \varepsilon_0$

$$\sigma(t) = \sigma_0 e^{\frac{-Et}{\eta}} = \varepsilon_0 E e^{\frac{-Et}{\eta}} = \varepsilon_0 E(t)$$

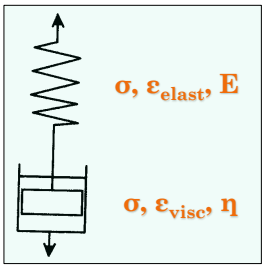
Relaxation modulus: $E(t) = Ee^{\frac{-Et}{\eta}}$

2.2 Behavior of Materials under Stress

Maxwell Model for Viscoelastic Behavior



relaxation



$$\sigma(t) = \varepsilon_0 E e^{\frac{-Et}{\eta}}$$

2.2 Behavior of Materials under Stress

Kelvin Model for Viscoelastic Behavior

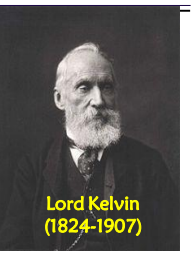
Assume stresses are additive
- Parallel System
- Both elements feel the same strain

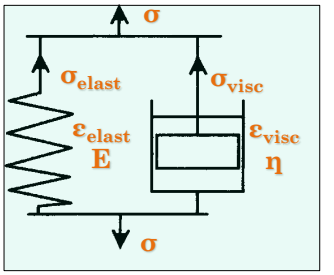
$$\varepsilon = \frac{1}{E}\sigma_1$$

$$\dot{\varepsilon} = \frac{1}{\eta}\sigma_2$$

$$\sigma = \sigma_1 + \sigma_2$$

$$\Rightarrow \quad \sigma = \varepsilon E + \eta \dot{\varepsilon}$$

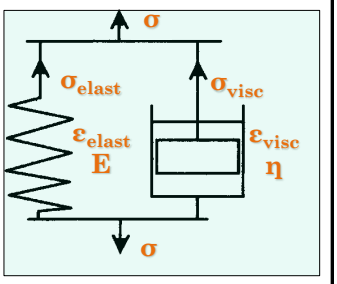


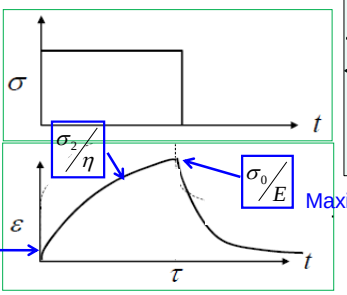


2.2 Behavior of Materials under Stress

Kelvin-Voigt Model for Viscoelastic Behavior

What is the response of this model to a sudden load σ_0?





Maximum strain

Initial slope: σ_0/η

$$\sigma = \varepsilon E + \eta \dot{\varepsilon}$$

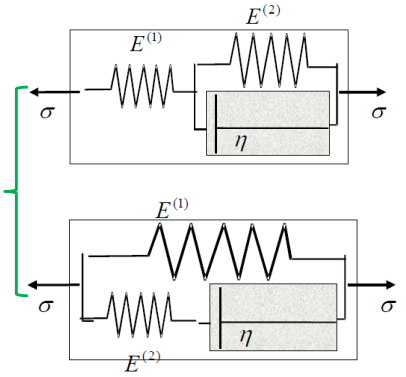
Initial condition: $\varepsilon(0) = 0$

$$\varepsilon(t) = \frac{\sigma_0}{E}\left(1 - e^{-\left(E/\eta\right)t}\right)$$

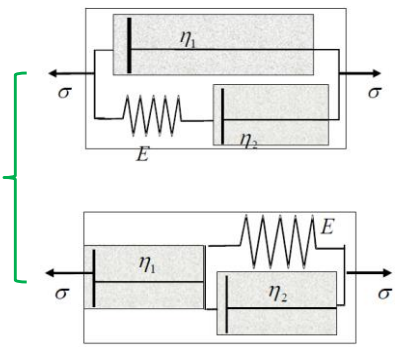
2.2 Behavior of Materials under Stress

Three-element Rheology Models

Standard solid



Standard fluid



2.2 Behavior of Materials under Stress

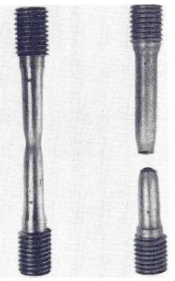
Plasticity:

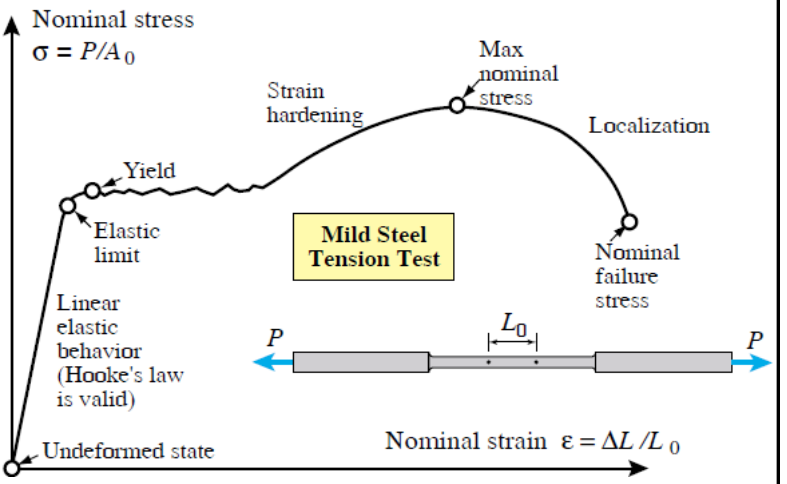




2.2 Behavior of Materials under Stress

Plasticity:





Typical tension test behavior of mild steel, which displays a well defined yield point and yield region

Summary

1. Physical Architecture of Materials

Atomic bonding: primary bonding; secondary bonding

Bonding energy: relationship with material properties

Colloid: concrete is a colloid

2. Behavior of Materials Under Stress

Modulus: tangent and secant modulus

Rheology: three basic elements; Maxwell and Kelvin models; creep and relaxation

3. Surface Properties

4. Failure and Fracture



QUESTIONS?